

Security and Privacy Considerations for Confirming Payments in Rwandan Mobile Money Systems



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Motivation

- 77% of Rwandan adults have a mobile money (MoMo) account; 56% rely solely on MoMo for banking
- How merchants confirm payments:

Intended practice: merchant and customer use own phones



Observed common practice: merchant relies on customer's phone



- Using **customers' phones** to confirm payment may lead to fraud (screenshots of past transactions, fake messages)

RQ1 How do Rwandan merchants verify MoMo payments from customers in practice?

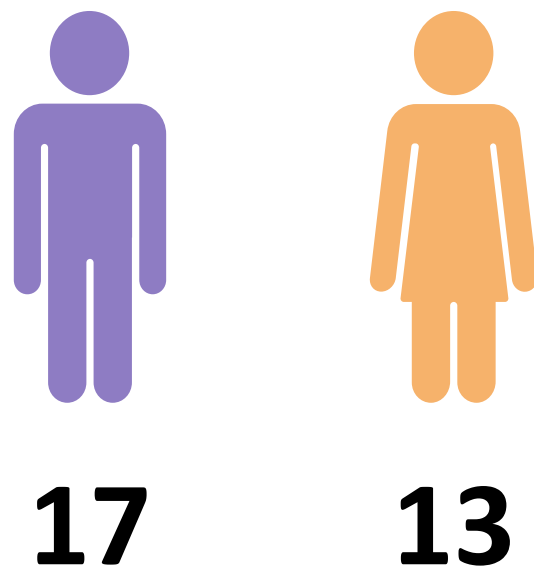
RQ2 What privacy concerns do users have when using current MoMo payment confirmation interfaces?

RQ3 What security vulnerabilities arise from current MoMo payment confirmation practices in Rwanda?

RQ4 How could MoMo payment confirmation workflows in Rwanda be redesigned for security and privacy?

Method: Semi-Structured Qualitative Interviews

Participants



- 28 Kinyarwanda
- 2 English

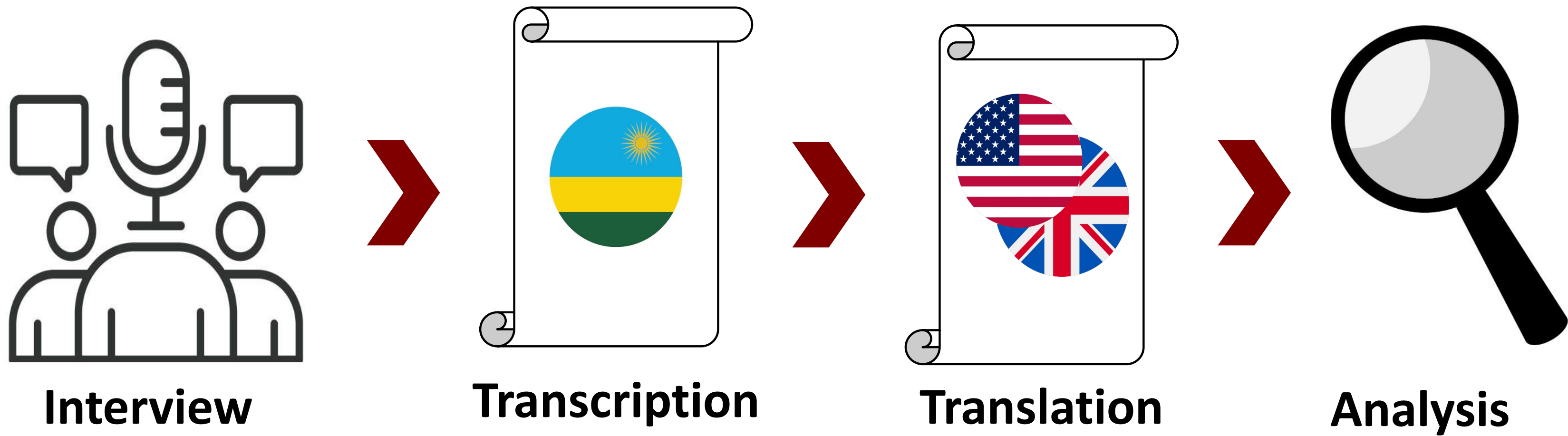
Recruitment Locations

- 18 Kigali City

- 12 Rural Areas



Data Analysis



Context

Customer's payment confirmation pop-up

MTN Rwandacell Message

Y'ello. You have paid 200 RWF to David, 188531. Fee 0 RWF. Transaction ID 32055055507. Your Mobile Money account balance is 386,386 RWF.

OK

Merchant's payment confirmation SMS

← M-Money ⋮

Today

You have received 200 RWF from Wallace (*****311) at 2025-09-13 18:53:35. Balance: 65214 RWF. FT Id: 20030504071. ET Id:-



Key Results

Payment Confirmation In Practice (RQ1a)

- 17/30 primarily used the customer's screen
 - 11 reported later verifying on own device
- 6/30 sometimes skipped payment confirmation entirely
- Some merchants looked at balance changes
- Ringtone and Vibrations
- Thumbs-up from customers

Reason For Using Customer's Phone (RQ1b)

- Network latency (delayed SMS)
- Device (e.g., owner's) not on premises
- Device in pocket / merchant wearing gloves
- Large volume of customers
- Established trust

Privacy Concerns (RQ2)

- Customer's balance shown on their screen
 - ".... I don't feel comfortable when the message also displays my account balance. I don't think merchants should see that." - P11
- Merchant's full name shown to customers

Security Concerns (RQ3)

- 24/30 had experienced fraud attempts
 - Reusing old screenshots
 - Shoulder surfing
 - Social engineering

Proposed Interface Redesigns (RQ4)

- Participants felt redesigns are needed to prevent fraud
- Positive reactions to splitting confirmations into two screens
- Reluctance to have customers hand over their phones
- Desire to specify multiple recipients of confirmation messages, but only if the store's balance is not included

1) Screen for Merchants

You have paid 1,200 RWF to:
David
(188531)
at 2025-09-13 18:51
TxID 32055055507
n Next

Cancel Send

2) Screen for Customers

Fee 0 RWF.
Your Mobile Money account balance
is now 386,386 RWF.

Cancel Send

Merchant-Defined Code

You have paid 1,200 RWF to:
David
(188531)
at 2025-09-13 18:51
TxID 32055055507

Verification code: 125730

Cancel Send



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